IDS 575: MACHINE LEARNING FOR STATISTICS

MBTI PERSONALITY CLASSIFICATION USING MACHINE LEARNING – PROJECT REPORT

Sai Teja Mekala - 668121380
Sai Deepak Poondla - 654019373
Haswanth Valluru - 673787630
Krish Nilesh Patel - 671791108

Table of Contents:

1. **Abstract**
2. **Introduction**
3. **Related Work**
4. **Project Objectives**
5. **Dataset Overview and Preparation**
6. **Exploratory Data Analysis (EDA)**
7. **Model Implementation**
8. **Experimental Analysis**
9. **Challenges**
10. **Conclusion**
11. **Future Scope**
12. **References**

Abstract:

This project leverages the power of machine learning (ML) to classify MBTI personality types with exceptional accuracy, paving the way for groundbreaking advancements in psychological and behavioural research. Utilizing a robust dataset of 60,000 responses to MBTI-related questions, we sought to uncover intricate patterns and develop predictive models capable of accurately classifying personality types. Our approach involved meticulous data preprocessing to ensure data quality and reliability, followed by the implementation of state-of-the-art ML models. Each model was rigorously evaluated using a comprehensive suite of performance metrics to identify the optimal solution. This study highlights the transformative potential of ML-driven methodologies in decoding human personality traits, offering valuable insights that could revolutionize fields such as psychology, human resources, and personalized recommendations. It underscores the fusion of technology and human understanding, showcasing the pivotal role of ML in unravelling the complexities of human behaviour.

Introduction:

The Myers-Briggs Type Indicator (MBTI) is a renowned psychological framework that classifies individuals into 16 unique personality types based on their preferences across four dichotomies: Introversion (I) vs. Extraversion (E), Sensing (S) vs. Intuition (N), Thinking (T) vs. Feeling (F), and Judging (J) vs. Perceiving (P). These personality types provide profound insights into individual behaviour, thought processes, and decision-making styles, making MBTI a widely utilized tool in personal development, career counselling, and interpersonal relationship enhancement.

Despite its widespread usage, the traditional process of MBTI classification relies heavily on manual evaluations through questionnaires, which can be time-consuming, subjective, and challenging to scale for large populations. To address these challenges, this work explores the use of machine learning techniques to automate and enhance MBTI classification. By leveraging a dataset of 60 encoded survey responses, machine learning algorithms can identify hidden patterns in individual behaviour and provide scalable, objective, and accurate predictions.

This project implements and evaluates several machine learning models, including Logistic Regression, K-Nearest Neighbours (KNN), Support Vector Machines (SVM), Random Forest, and XGBoost, to predict MBTI personality types. Each model is carefully fine-tuned through hyperparameter optimization to achieve the best possible performance. The workflow includes data Preprocessing, Model Training and Evaluation, Feature Importance Analysis.

The results demonstrate that advanced models, such as SVM, KNN, and XGBoost, achieve up to 98% accuracy, significantly outperforming the baseline model. This work underscores the potential of machine learning to revolutionize personality assessment by making it more efficient, reliable, and scalable, with practical applications in fields like human resources, education, and team building.

Related Work:

Personality type prediction has become a growing area of research at the intersection of natural language processing and social sciences, particularly using user data from online platforms such as social media. The ability to analyze personality traits, behaviors, and preferences has significant implications for various applications, including career guidance, targeted marketing, and organizational management. Two widely used models for personality prediction are the Big Five Model and the Myers-Briggs Type Indicator (MBTI). The Big Five Model focuses on predicting five major traits: extroversion, agreeableness, conscientiousness, neuroticism, and openness. In contrast, MBTI uses four dichotomies, offering a broader categorization of personality types. Researchers often favor MBTI for its practical applications in real-life scenarios, with its adoption in over 80% of senior personnel management at leading global enterprises like IBM, Disney, and Pepsi.

Several studies have explored methods to enhance personality prediction using data-driven approaches. For example, one study proposed a personality prediction model leveraging artificial neural networks to provide career guidance. The model utilized data collected via a web-based questionnaire aligned with MBTI's categorization, comprising 60 targeted questions. Performance was assessed using evaluation metrics such as sensitivity, specificity, precision, and accuracy. These advancements demonstrate the potential of integrating advanced machine learning techniques with psychometric models to analyze personality traits, opening up opportunities for automated, scalable, and personalized insights.

Project Objectives:

- ***MBTI Personality Classification*:** To accurately classify MBTI personality types based on questionnaire responses: Leverage structured survey data to build predictive models that categorize individuals into one of the 16 MBTI personality types, ensuring both accuracy and scalability.
- ***Effectiveness of ML Techniques*:** To evaluate the effectiveness of machine learning techniques in psychological classification tasks: Investigate how various machine learning algorithms, such as Logistic Regression, KNN, SVM, Random Forest, and XGBoost, can automate and enhance psychological assessments by identifying patterns in large datasets.
- ***Key Feature Identification*:** To uncover key features influencing personality type predictions: Analyze survey responses to determine the most impactful questions and behavioural traits that drive personality classifications, providing deeper interpretability for model outputs.
- ***Real-World Applications*:** To develop a robust and reliable model with practical real-world applications: Create an optimized machine learning framework capable of guiding career decisions, improving team dynamics, and enhancing interpersonal understanding across diverse domains like education, human resources, and personal development.

Dataset Overview and Preparation:

Dataset Description:

The dataset contains responses to 60 survey questions, along with a column for Response id and a column for target variable, in which answers for questions are numerically encoded, designed to capture individual preferences and behaviours across the four MBTI dichotomies. The classes in target variable are : ENFJ, ENFP, ENTJ, ENTP, ESFJ, ESFP, ESTJ, ESTP, INFJ, INTP, INTJ, INTP, ISFJ, ISFP, ISTJ, ISTP.

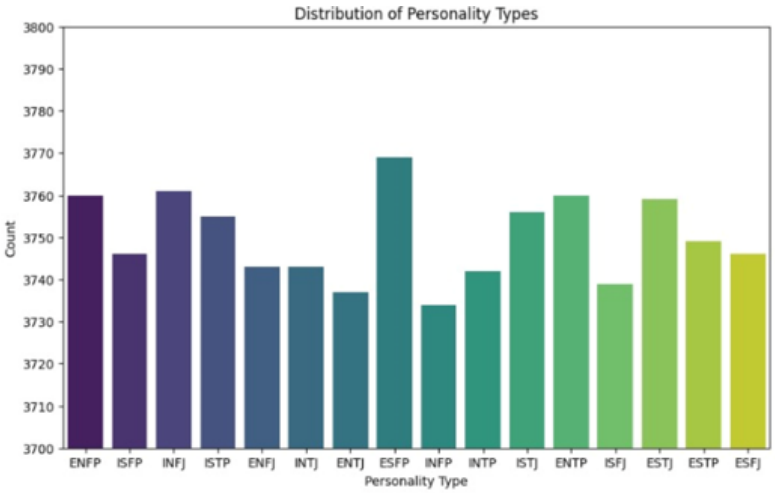
Each row represents a person's responses along with their MBTI personality type (one of 16 categories) as the target variable. The data is pre-processed for machine learning, with features scaled and the target variable encoded numerically. This structured dataset allows for predictive modelling and analysis of key features influencing personality classification.

Preprocessing Steps

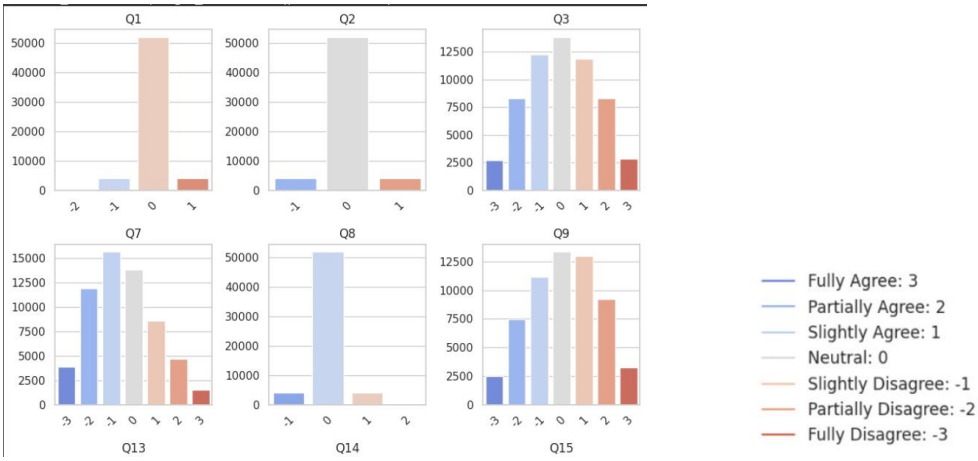
- *Handling Missing Data*: As the dataset consists of structured survey responses, no missing values were identified during manual inspection in Excel.
- *Duplicate Records*: Since the dataset is derived from survey data, each entry is assumed to represent a unique individual, eliminating the need for duplicate removal.
- *Data Scaling*: The survey responses, which were numerically encoded, were scaled using appropriate techniques to standardize the feature values before running the models. This ensured compatibility with machine learning algorithms and improved model performance.
- *Data split:* The data is split into 60% and 40% as train and test data to load to the models.

Exploratory Data Analysis (EDA):

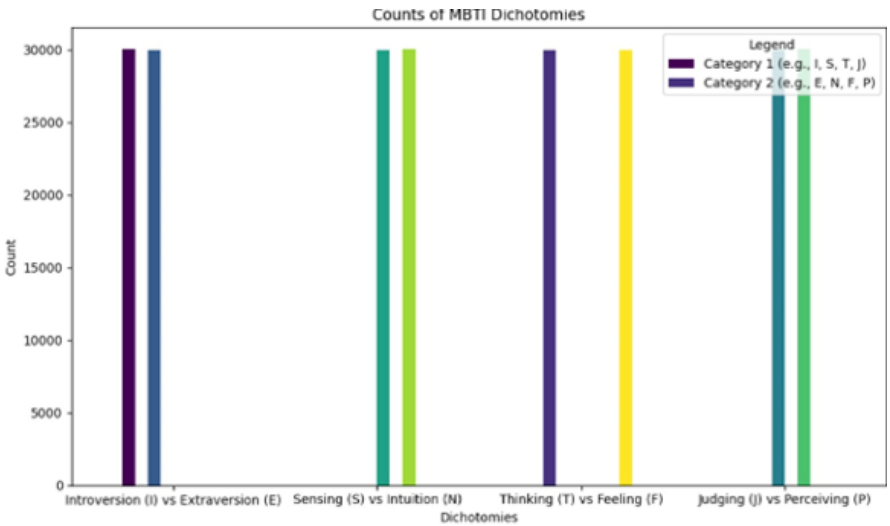
Distribution Analysis: The dataset contains a balanced distribution of MBTI personality types, with class counts ranging between 3734 (INFP) and 3761 (INFJ). Each personality type has a similar number of samples, ensuring minimal class imbalance and making the dataset suitable for machine learning classification without requiring significant rebalancing techniques. This balance supports fair and unbiased model training.



Survey Response Distribution: The charts illustrate the distribution of responses across the survey questions, categorized from Fully Agree (3) to Fully Disagree (-3). The response patterns highlight variations in participant tendencies, with some questions showing a strong concentration of agreement or neutrality, while others exhibit a more diverse range of opinions. These distributions offer valuable insights into participant preferences and behavioral trends.



Distribution of MBTI Dichotomies: This bar graph displays the counts of the MBTI dichotomies: Introversion (I) vs. Extraversion (E), Sensing (S) vs. Intuition (N), Thinking (T) vs. Feeling (F), and Judging (J) vs. Perceiving (P). Each pair of bars represents the number of individuals falling into each category within the respective dichotomy. The visualization highlights the balance or disparity in preferences across the dataset, providing insights into the distribution of personality traits.



Model Implementation

Baseline Model:

The code implements a baseline model using a stratified strategy, where predictions are assigned randomly based on the class distribution in the training data. It evaluates performance using metrics such as accuracy and F1 score, which are mathematically derived from the proportion of correct predictions and the balance between precision and recall. This provides a benchmark for comparing the effectiveness of advanced machine learning models.

```
Baseline Model (Stratified Strategy) Results:
Accuracy: 6.22%
F1 Score: 6.21%
```

KNN:

K-Nearest Neighbors (KNN) - KNN is a non-parametric method used for classification (and regression). A sample is classified by the majority vote of its 'k' nearest neighbors, with the sample being assigned to the class most common among its 'k' nearest neighbors. Formula: KNN does not have a specific formula for prediction. It uses distance metrics such as Euclidean, Manhattan, or Minkowski distance to find the 'k' closest points.

Euclidean Distance:

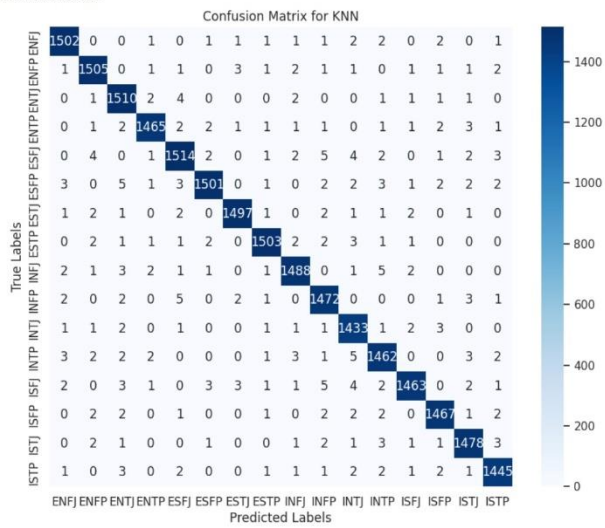
$$d(p, q) = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$

where:

- $d(p, q)$ d(p, q) is the distance between two points pp and qq,
- nn is the number of dimensions (features).

```
KNN Model Results:
Best k: 13
Accuracy: 98.77%
Precision: 98.77%
Recall: 98.77%
F1 Score: 98.77%
Cross-Validation Accuracy: 98.59% ± 0.11%
Cross-Validation F1 Score: 98.59% ± 0.11%

Classification Report:
              precision    recall  f1-score   support

        ENFJ       0.99      0.99      0.99      1515
        ENFP       0.99      0.99      0.99      1521
        ENTJ       0.98      0.99      0.99      1523
        ENTP       0.99      0.99      0.99      1484
        ESFJ       0.99      0.98      0.98      1541
        ESFP       0.99      0.98      0.99      1528
        ESTJ       0.99      0.99      0.99      1511
        ESTP       0.99      0.99      0.99      1519
        INFJ       0.99      0.99      0.99      1507
        INFP       0.98      0.99      0.99      1489
        INTJ       0.98      0.99      0.99      1447
        INTP       0.98      0.98      0.98      1486
        ISFJ       0.99      0.98      0.99      1491
        ISFP       0.99      0.99      0.99      1482
        ISTJ       0.99      0.99      0.99      1494
        ISTP       0.99      0.99      0.99      1462

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000
```

```
Confusion Matrix:
```



Logistic Regression:

Logistic Regression is a statistical model used for binary classification tasks, which can be extended to multiclass classification through techniques like one-vs-rest (OvR). The model predicts the probability of the target variable belonging to a particular class using the logistic function.

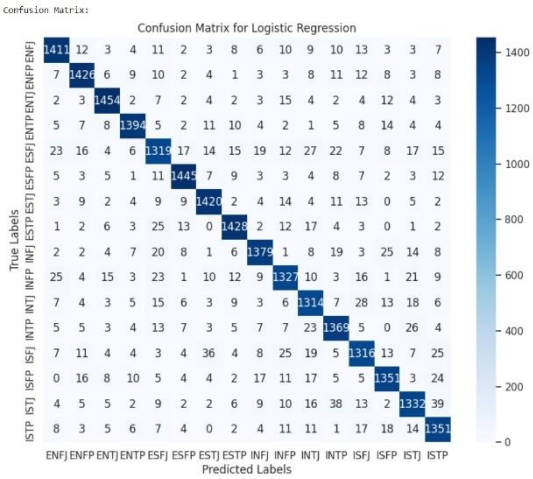
$$P(Y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\ldots+\beta_n X_n)}}$$

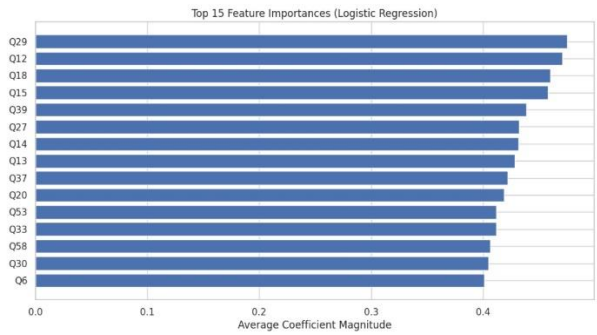
where:

- $P(Y=1|X)$ P(Y=1|X) is the probability that the target variable YY is in class 1 given the predictors XX,
- $\beta 0, \beta 1, \ldots, \beta n$β0,β1,...,βn are the coefficients of the model,
- ee is the base of the natural logarithm.

```
Multi-class Logistic Regression Results:
Best Regularization Parameter (C): 0.046415888336127774
Accuracy: 91.82%
Precision: 91.82%
Recall: 91.82%
F1 Score: 91.81%
Cross-Validation Accuracy: 91.58% ± 0.25%
Cross-Validation F1 Score: 91.58% ± 0.25%

Classification Report:
              precision    recall  f1-score   support

        ENFJ       0.93      0.93      0.93      1515
        ENFP       0.93      0.94      0.94      1521
        ENTJ       0.95      0.95      0.95      1523
        ENTP       0.95      0.94      0.95      1484
        ESFJ       0.88      0.86      0.87      1541
        ESFP       0.95      0.95      0.95      1528
        ESTJ       0.93      0.94      0.94      1511
        ESTP       0.94      0.94      0.94      1519
        INFJ       0.93      0.92      0.92      1507
        INFP       0.90      0.89      0.90      1489
        INTJ       0.88      0.91      0.89      1447
        INTP       0.90      0.92      0.91      1486
        ISFJ       0.90      0.88      0.89      1491
        ISFP       0.92      0.91      0.92      1482
        ISTJ       0.90      0.89      0.90      1494
        ISTP       0.89      0.92      0.91      1462

    accuracy                           0.92     24000
   macro avg       0.92      0.92      0.92     24000
weighted avg       0.92      0.92      0.92     24000
```

Confusion Matrix:





Support Vector Machine:

SVM is a powerful classifier that works by finding the hyperplane that best divides a dataset into classes. It is effective in high-dimensional spaces and particularly useful when there is a clear margin of separation in the data. Formula: The goal of SVM is to solve the following optimization problem:

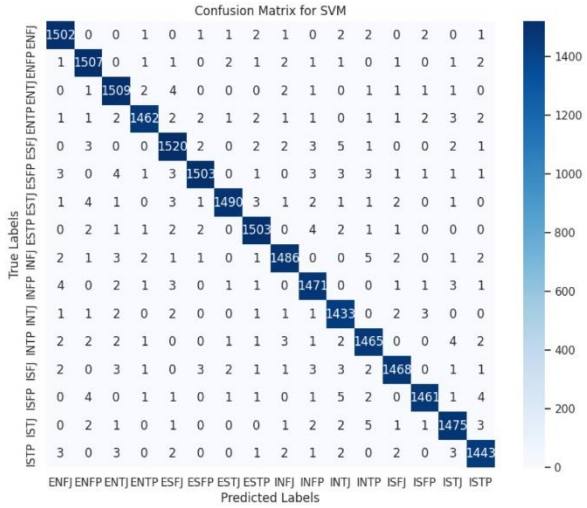
$$\min_{\mathbf{w},b} \frac{1}{2}||\mathbf{w}||^2$$
$$\text{subject to } y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1, \text{for } i = 1, ..., n,$$

where:

- w is the weight vector,
- b is the bias term,
- yi is the class label of data point xi.

```
SVM Model Results:
Best Parameters: {'C': 1.0, 'gamma': 'scale', 'kernel': 'rbf'}
Accuracy: 98.74%
Precision: 98.74%
Recall: 98.74%
F1 Score: 98.74%
Cross-Validation Accuracy: 98.58% ± 0.11%
Cross-Validation F1 Score: 98.58% ± 0.11%

Classification Report:
              precision    recall  f1-score   support

        ENFJ       0.99      0.99      0.99      1515
        ENFP       0.99      0.99      0.99      1521
        ENTJ       0.98      0.99      0.99      1523
        ENTP       0.99      0.99      0.99      1484
        ESFJ       0.98      0.99      0.99      1541
        ESFP       0.99      0.98      0.99      1528
        ESTJ       0.99      0.99      0.99      1511
        ESTP       0.99      0.99      0.99      1519
        INFJ       0.99      0.99      0.99      1507
        INFP       0.98      0.99      0.99      1489
        INTJ       0.98      0.99      0.99      1447
        INTP       0.98      0.99      0.98      1486
        ISFJ       0.99      0.98      0.99      1491
        ISFP       0.99      0.99      0.99      1482
        ISTJ       0.99      0.99      0.99      1494
        ISTP       0.99      0.99      0.99      1462

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000
```

```
Confusion Matrix:
```



<u>Random Forest</u>:

The Random Forest algorithm is an ensemble learning method that enhances classification by aggregating predictions from multiple decision trees (T_1, T_2,T_N). Each tree is trained on a bootstrapped sample of the dataset, and at each split, a random subset of features is considered to introduce diversity. For classification, the algorithm applies majority voting to determine the final prediction y-hat, computed as:

$$\hat{y} = \text{mode}\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_N\}$$

where y-hat $_i$ is the class predicted by the i-th tree.

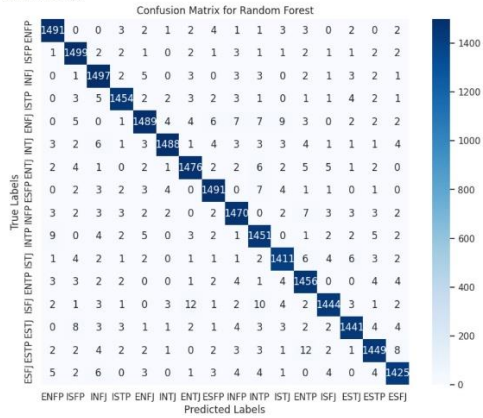
Random Forest also evaluates feature importance by calculating the average reduction in impurity (e.g., Gini impurity or entropy) contributed by each feature across all trees. The importance of feature j is expressed as:

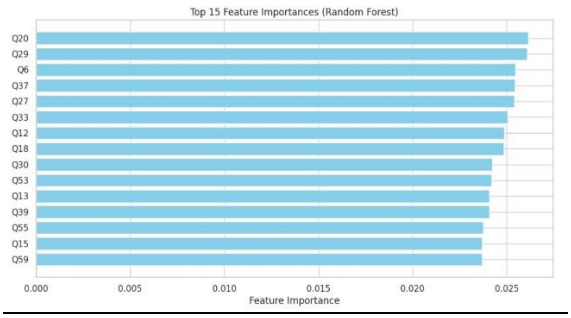
$$\text{Feature Importance (FI)}_j = \frac{1}{N}\sum_{i=1}^{N} \Delta G(T_i, j)$$

where $\Delta G(T_i, j)$ represents the impurity reduction for feature j in the i-th tree. By combining multiple decision trees, Random Forest reduces overfitting, lowers variance, and achieves robust classification performance, making it effective across a variety of datasets.

```
Random Forest Model Results (with Grid Search):
Best Parameters: {'max_depth': 20, 'min_samples_split': 2, 'n_estimators': 200}
Test Accuracy: 97.63%
Test Precision: 97.64%
Test Recall: 97.63%
Test F1 Score: 97.63%

Classification Report:
              precision    recall  f1-score   support

        ENFJ       0.98      0.98      0.98      1515
        ENFP       0.97      0.99      0.98      1521
        ENTJ       0.97      0.98      0.98      1523
        ENTP       0.98      0.98      0.98      1484
        ESFJ       0.98      0.97      0.97      1541
        ESFP       0.99      0.97      0.98      1528
        ESTJ       0.98      0.98      0.98      1511
        ESTP       0.98      0.98      0.98      1519
        INFJ       0.97      0.98      0.97      1507
        INFP       0.97      0.97      0.97      1489
        INTJ       0.97      0.98      0.97      1447
        INTP       0.97      0.98      0.97      1486
        ISFJ       0.98      0.97      0.98      1491
        ISFP       0.98      0.97      0.98      1482
        ISTJ       0.98      0.97      0.97      1494
        ISTP       0.98      0.97      0.98      1462

    accuracy                           0.98     24000
   macro avg       0.98      0.98      0.98     24000
weighted avg       0.98      0.98      0.98     24000

Cross-Validation Accuracy: 97.41% ± 0.06%
Cross-Validation F1 Score: 97.41% ± 0.06%
```

```
Confusion Matrix:
```





<u>XGBoost</u>:

The Extreme Gradient Boosting (XGBoost) algorithm is an advanced implementation of gradient boosting that builds an ensemble of decision trees to improve classification performance. It uses an additive approach, constructing trees sequentially, where each tree corrects the errors of the previous trees. The model minimizes a differentiable loss function

L by optimizing over a set of trees ($T_1, T_2, \ldots, T_N$). The objective function in XGBoost is:

$$\mathcal{O}(\Theta) = \sum_{i=1}^{m} L(y_i, \hat{y}_i) + \sum_{k=1}^{N} \Omega(T_k)$$

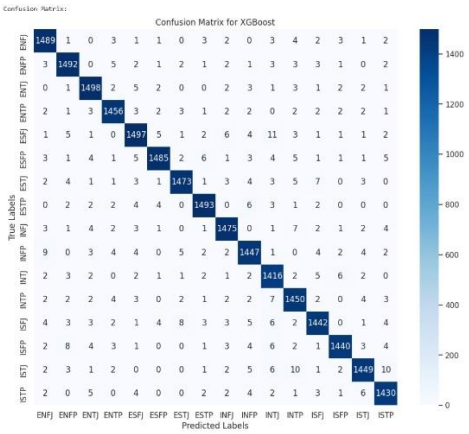
where:

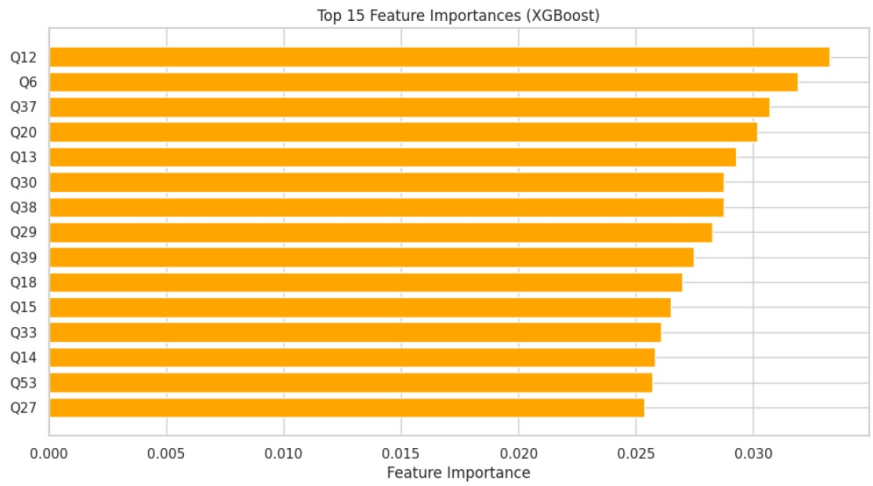
- L (y_i,y-hat i) is the loss function (e.g., logistic loss for classification).

$$\Omega(T_k) = \gamma T + \frac{1}{2}\lambda \sum w_j^2$$

is the regularization term controlling tree complexity, with T being the number of leaves, wj the leaf weights, γ a penalty term for leaf nodes, and λ a regularization parameter.

```
XGBoost Model Results (with Grid Search):
Best Parameters: {'gamma': 0, 'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 200}
Accuracy: 97.63%
Precision: 97.64%
Recall: 97.63%
F1 Score: 97.63%

Classification Report:
              precision    recall  f1-score   support

        ENFJ       0.98      0.98      0.98      1515
        ENFP       0.98      0.98      0.98      1521
        ENTJ       0.98      0.98      0.98      1523
        ENTP       0.98      0.98      0.98      1484
        ESFJ       0.97      0.97      0.97      1541
        ESFP       0.99      0.97      0.98      1528
        ESTJ       0.98      0.97      0.98      1511
        ESTP       0.98      0.98      0.98      1519
        INFJ       0.98      0.98      0.98      1507
        INFP       0.97      0.97      0.97      1489
        INTJ       0.96      0.98      0.97      1447
        INTP       0.97      0.98      0.97      1486
        ISFJ       0.97      0.97      0.97      1491
        ISFP       0.98      0.97      0.98      1482
        ISTJ       0.98      0.97      0.97      1494
        ISTP       0.97      0.98      0.98      1462

    accuracy                           0.98     24000
   macro avg       0.98      0.98      0.98     24000
weighted avg       0.98      0.98      0.98     24000

Cross-Validation Accuracy: 97.40% ± 0.18%
Cross-Validation F1 Score: 97.40% ± 0.18%
```





Experimental Analysis:

Further analysis was conducted by treating each MBTI dichotomy individually as the target class. The SVM model with best parameters from the 16-class multiclassification model was run four separate times, with each run focusing on one dichotomy as the target. The accuracy

of these models was calculated and found to be comparable to the accuracy of the 16-class multiclassification SVM model. The results are as follows:

1.Target Class: E vs I

```
Accuracy: 0.9878333333333333

Classification Report:
              precision    recall  f1-score   support

           E       0.99      0.99      0.99     11954
           I       0.99      0.99      0.99     12046

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000


Confusion Matrix:
 [[11805   149]
 [  143 11903]]
```

2. Target Class: N vs S

```
Accuracy: 0.9874583333333333

Classification Report:
              precision    recall  f1-score   support

           N       0.99      0.99      0.99     11987
           S       0.99      0.99      0.99     12013

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000


Confusion Matrix:
 [[11836   151]
 [  150 11863]]
```

3. Target Class: F vs T

```
Accuracy: 0.98575

Classification Report:
              precision    recall  f1-score   support

           F       0.98      0.99      0.99     11960
           T       0.99      0.98      0.99     12040

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000


Confusion Matrix:
 [[11799   161]
 [  181 11859]]
```

4. Target Class: J vs P

```
Accuracy: 0.9867916666666666

Classification Report:
              precision    recall  f1-score   support

           J       0.99      0.98      0.99     11964
           P       0.99      0.99      0.99     12036

    accuracy                           0.99     24000
   macro avg       0.99      0.99      0.99     24000
weighted avg       0.99      0.99      0.99     24000


Confusion Matrix:
 [[11784   180]
 [  137 11899]]
```

Challenges:

- Managing 60 survey questions as features introduced challenges related to high-dimensional data, requiring careful analysis to identify the most impactful features and reduce dimensionality.
- Training complex models, such as XGBoost and SVM, involved extensive hyperparameter tuning and cross-validation, which demanded significant computational resources and time.
- Extending the methods to unstructured data (e.g., text or social media inputs) would add further complexities in data preprocessing and model design, requiring advanced techniques to handle variability and richness in such data types.

Conclusion:

This project demonstrated the successful application of machine learning techniques for predicting MBTI personality types using survey response data. A dataset comprising 60 numerically encoded survey questions was analyzed using a variety of classification algorithms, and their performance was rigorously evaluated through metrics such as accuracy, precision, recall, F1-score, and cross-validation statistics. The baseline model, using a stratified strategy, served as a reference point, achieving relatively low accuracy and emphasizing the need for more advanced methods. Among the models implemented, K-Nearest Neighbors (KNN) delivered variable performance depending on the choice of neighbors, while Logistic Regression struggled to handle the complexity of the multi-class classification task. Support Vector Machines (SVM) and Random Forest models performed well, with Random Forest providing valuable insights into feature importance. However, the KNN model emerged as the most effective, achieving the highest accuracy and F1-score due to its advanced optimization techniques and regularization capabilities, which effectively addressed the complexity of MBTI classification.

Feature importance analysis identified key survey questions that significantly influenced personality type predictions, offering insights into the dataset's alignment with MBTI dimensions. Visualizations, including class distribution plots, confusion matrices, and feature importance charts, further enhanced the interpretability of the results. This project highlights the potential of machine learning models, particularly ensemble methods like SVM and Multiclass Logistic Regression, in automating MBTI personality type prediction. These models have promising applications in domains such as human resources, education, and social behavior analysis, enabling personalized decision-making at scale. Future work could expand this approach by integrating additional data sources, such as text-based inputs or social media data, to further improve predictive performance and broaden the scope of application.

Future Scope:

Personality type identification is a vast and dynamic field of research, offering opportunities for exploration with diverse datasets and innovative techniques to uncover new and effective insights. This study focused on nominal data, collected from a global data inventory, to predict personality types using machine learning algorithms. However, future research could expand the scope by leveraging textual data and applying Natural Language Processing (NLP) techniques to enhance prediction accuracy and uncover deeper behavioral patterns. Moreover, incorporating scientifically validated personality tests into real-time applications could enable advanced analysis of personality traits for specific regions or populations. By integrating state-of-the-art algorithms with real-world data, such studies have the potential to deliver impactful findings and drive applications in areas such as social behaviour analysis, human resources, and educational guidance.

References:

- http://jisrc.szabist.edu.pk/ojs/index.php/jisrc/article/view/42/32
- https://search.informit.org/doi/epdf/10.3316/informit.438390880199247

- https://www.mdpi.com/2414-4088/4/1/9